CBE 176 : MT#1 Cheat Sheet

Red Cat → Reduction @ Cathode ($e^- \rightarrow$)

An Ox → Oxidation @ Anode ($\rightarrow e^-$)

$F = 96485 \ \frac{C}{mol}$

e^- flow from: Low potential → high potential

High free energy → Low free energy

Electroneutrality

- $i A_{Anode} + i A_{Cathode} = 0$ (current density $\left[\frac{mA}{cm^2}\right]$; surface area)
- $\sum_i z_i c_i = 0$ (charge on species; conc. of species)
- $i = F \sum z_i N_i = F(z_+ N_+ + z_- N_-)$ (Flux [- Diffusion, - Migration, - Convection])

Total mass consumed/formed

- $M_i = \frac{s_i (MW_i) Q}{nF}$; $Q = \int I(t)\,dt = I \cdot t$ [-] mAh

[Sign convention written as reduction rxn]

- Total $e^- = \frac{Q}{F} (6.023 \times 10^{23})$; $W = \int I(t) V(t)\,dt$
- Faradaic efficiency: $\frac{Q_{desired}}{Q_{total}} \times 100$

Thermodynamics

- Equilibrium / Open-circuit potential

$\Delta G^\circ_{rxn} = -nFU^\circ_{rxn}$ [$U^\circ > 0$, $\Delta G^\circ < 0$ → spontaneous]

$\bar{\mu}_i = \mu_i + z_i F \underline{\Phi}$; $\mu_i = f(N,T,P)$; $\bar{\mu}_{e^-} = -F\underline{\Phi} = -FU^\circ$

$\mu_{e^-} = 0$

$\bar{\mu}_i = \mu_i$ [for an uncharged specis]

$\eta = \underline{\Phi} - U$; $\underline{\Phi} > U$ → decreases energy of e^- (when positive potential applied)

→ e^- transfer from reactants to electrode

→ Anodic Process / oxidation

$V_{cell} = U_{right} - U_{left}$ @ equilibrium [$V_{cell} > 0$ for galvanic cell]

- Phase Equilibrium



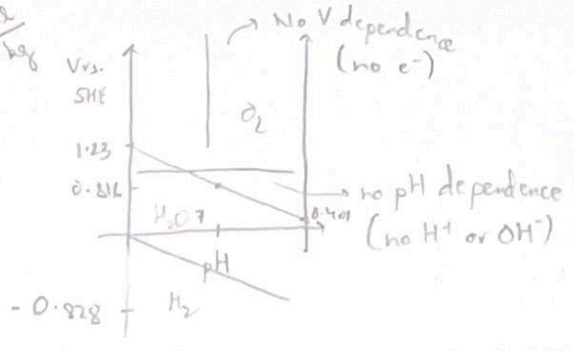
$\bar{\mu}_i^{\alpha} = \bar{\mu}_i^{\beta}$ [Adjacent phases have equal (electro)chemical potentials of the same specis 'i'] } same specis diff phase rule

$U = \underline{\Phi}^{\alpha'} - \underline{\Phi}^{\alpha}$

$FU = \mu_{e^-}^{\alpha} - \mu_{e^-}^{\alpha'}$ [Left - Right]

μ are G_f values from NBS table @ 1 bar, 25°C, 1 mol/kg

For a strong electrolyte → we assume complete dissociation & therefore can use starting specis.



$M \leftrightarrow M^{n+} + ne^-$ [Horizontal line]

$M + nH_2O \leftrightarrow M(OH)_n + nH^+ + ne^-$

$M + nH_2O \leftrightarrow MO_n + 2nH^+ + ne^-$ } slanted

$M + nH_2O \leftrightarrow MO_n^{n-} + nH^+ + ne^-$ [Vertical]

· Kinetics $i = i_0 \left[\exp\left(\frac{\alpha_A}{RT} F \eta_s \right) - \exp\left(-\frac{\alpha_c}{RT} F \eta_s \right) \right]$ $\alpha_A = 1 - \alpha_c$

$\vec{i} = -K \nabla \Phi$; We want high K, high current, low η

· Deviations from 1 mol/kg molalities

↳ Non-ionizing solutes / Weak electrolytes: (m_i → molality of i)

$$\mu_i = \mu_i^\circ + RT \ln\left(\frac{m_i}{m_\circ}\right) + RT \ln(\gamma_i)$$

↳ Ionizing solutes / strong electrolytes:

$$\mu_i = \mu_i^\circ + \nu RT \ln(m_A \gamma_{1-}) + RT \ln(\nu_+^{\nu_+} \nu_-^{\nu_-}) \quad ; \quad \nu = \nu_+ + \nu_-$$

↳ Solvents: (ϕ → osmotic coefficient)

$$\mu_0 = \mu_0^\circ - \nu (MW) RT m_A \phi$$

↳ Gases:

$$\mu_A = \mu_A^\circ + RT \ln(P_A) \quad ; \quad P_A = x_A P f_A$$

↳ Solids:

(Pure) $\mu_A = \mu_A^\circ$

(Alloys) $\mu_A = \mu_A^\circ + RT \ln a_A$ (a_A → activity of A in alloy.)

· Deviations from T & P:

Pressure: $U - U^\circ = -\frac{2.303\, RT}{nF} \log\left(\frac{P}{P^\circ}\right)$ @ n=1, 25C ⇒ $U - U^\circ = 60\,mV \log\left(\frac{P}{P^\circ}\right)$

$\ln(x) = 2.303 \log(x)$

Temp: $\Delta H = \Delta G - T\Delta S = -T^2 \frac{\partial\left(\frac{\Delta G}{T}\right)}{\partial T}$ [Gibbs Helmholtz]

Closed system: $\Delta U = \frac{S^\circ}{nF}(T - T^\circ)$ → small temp. changes.

· pH vs. V

Nernst: $U = U^\circ - \frac{RT}{nF} \ln\left(\Pi m_i^{s_i}\right)$; $U = U^\circ - \frac{0.05916}{n} \log\left[\Pi \frac{m_{reduced}^{s_i}}{m_{oxidized}^{s_i}}\right]$ @ T = 25°C

CBE 124: MT#1 Cheat Sheet

F = 96485 C/e⁻

Nernst-Planck: $\frac{dC_i}{dt} = -\nabla \cdot \left[\underbrace{-z_i U_i F C_i \nabla \Phi + C_i \vec{v} - D_i \nabla C_i}_{N_i} \right] + R_i$ → For strong electrolytes: $\frac{\partial c}{\partial t} = D \nabla^2 c - \vec{v} \cdot \nabla c$

$\vec{i} = F \sum z_i \vec{N}_i$

$\sum z_i i_i = 0$

Conductive diffusion: $i = -\nabla \Phi \cdot \underbrace{\left[F^2 \sum z_i^2 U_i c_i \right]}_{\kappa} - F \sum z_i D_i \nabla c_i \quad ; \quad \kappa = z_+ c_+ (\lambda_+ + \lambda_-)$

↳ $\nabla c = 0$: $i = -\kappa \nabla \Phi$; $R = \frac{\rho l}{A}$; $\kappa = \frac{1}{s}$

↳ $\nabla c \neq 0$: $\nabla \Phi = \frac{-i}{\kappa} - \underbrace{\frac{F}{\kappa} \sum z_i D_i c_i}_{\text{Diffusion potential}}$

z = charge of species c_i

Nernst Einstein relationship: $D_i = RT U_i = \frac{z_+ U_+ D_- - z_- U_- D_+}{z_+ U_+ - z_- U_-} = \frac{D_+ D_- (z_+ - z_-)}{z_+ D_+ - z_- D_-}$

Transference number: $t_+ = \frac{z_i F \vec{N}_i}{\vec{i}} = \frac{z_i^2 U_i c_i}{\sum_j z_j^2 U_j c_j} = \frac{z_+ D_+}{z_+ D_+ - z_- D_-} = 1 - t_-$

Ion Diffusion: $D_+ = \frac{RT}{(\pi \eta r}$; $D_+ \propto \frac{1}{\eta}$; $D_+ \propto \frac{1}{r}$

